

ECN 594: Midterm Review Notes

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1 Demand Estimation

1.1 Why Demand Estimation?

- We need demand models to:
 - Measure substitution patterns between products
 - Compute price elasticities
 - Evaluate policy (mergers, new products, price changes)
 - Calculate consumer welfare
- The dimensionality problem: J products $\rightarrow J^2$ elasticities
- Solution: characteristics-based models (Lancaster, BLP)

1.2 The Logit Model: Key Equations

- Utility: $u_{ijt} = \delta_{jt} + \varepsilon_{ijt}$ where $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$
- Choice probability (market share):

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^J \exp(\delta_{kt})}$$

- Outside option: $s_{0t} = \frac{1}{1 + \sum_{k=1}^J \exp(\delta_{kt})}$

- **Berry inversion:**

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$$

1.3 Logit Elasticities

- **Own-price elasticity:**

$$\eta_{jjt} = \frac{\partial s_{jt}}{\partial p_{jt}} \cdot \frac{p_{jt}}{s_{jt}} = \alpha p_{jt}(1 - s_{jt})$$

- **Cross-price elasticity:**

$$\eta_{jkt} = \frac{\partial s_{jt}}{\partial p_{kt}} \cdot \frac{p_{kt}}{s_{jt}} = -\alpha p_{kt}s_{kt}$$

- Note: $\alpha < 0$ so own-price elasticity is negative
- **IIA problem:** Cross-elasticity depends only on k , not on similarity to j

1.4 The Identification Problem

- **Problem:** We observe equilibrium (p, q) pairs
- Can't tell if demand shifted or supply shifted
- **Price endogeneity:** High unobserved quality $\xi_{jt} \rightarrow$ high price
 - OLS sees: high price, still high demand
 - Concludes: price doesn't matter much
 - Result: $\hat{\alpha}$ biased toward zero
- **Solution:** Instrumental variables
 - Need: correlated with price, uncorrelated with ξ
 - Examples: Hausman IVs, BLP IVs, cost shifters

1.5 Consumer Surplus: Log-Sum Formula

- Expected utility for consumer i :

$$E[\max_j u_{ijt}] = \ln \left[\sum_{j=0}^J \exp(\delta_{jt}) \right] + \text{constant}$$

- Consumer surplus (in dollars):

$$CS = \frac{1}{|\alpha|} \ln \left[\sum_{j=0}^J \exp(\delta_{jt}) \right]$$

- Divide by $|\alpha|$ to convert utils to dollars
- **Application:** Welfare change from adding/removing products

1.6 The IIA Problem

- **IIA:** Ratio of choice probabilities doesn't depend on other options

$$\frac{s_{jt}}{s_{kt}} = \frac{\exp(\delta_{jt})}{\exp(\delta_{kt})} = \exp(\delta_{jt} - \delta_{kt})$$

- **Red Bus / Blue Bus:** Adding identical bus option incorrectly increases welfare
- **Problem:** Logit doesn't know similar products are close substitutes
- **Solutions:**
 - Demographic interactions (partial fix)
 - Mixed logit / random coefficients (full fix, beyond scope)

1.7 Quick Summary: Logit Formulas

Formula	Expression
Own-price elasticity	$\eta_{jkt} = \alpha p_{jt}(1 - s_{jt})$
Cross-price elasticity	$\eta_{jkt} = -\alpha p_{kt}s_{kt}$
Berry inversion	$\delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$
Consumer surplus	$CS = \frac{1}{ \alpha } \ln \left[\sum_j \exp(\delta_{jt}) \right]$

Remember: $\alpha < 0$, so own-price elasticity is negative.

2 Practice Problems: Demand Estimation

2.1 Practice: Elasticity Calculation

Question: Suppose $\alpha = -0.5$, product A has price $p_A = 20$ and share $s_A = 0.1$.

- (a) Calculate the own-price elasticity of product A.
- (b) If product B has $p_B = 25$ and $s_B = 0.15$, what is the cross-price elasticity of A with respect to B's price?

Solution:

- (a) **Own-price elasticity:**

$$\eta_{AA} = \alpha p_A(1 - s_A) = (-0.5)(20)(1 - 0.1) = -9$$

Demand is elastic (a 1% price increase reduces quantity by 9%)

- (b) **Cross-price elasticity:**

$$\eta_{AB} = -\alpha p_B s_B = -(-0.5)(25)(0.15) = 1.875$$

A 1% increase in B's price increases A's share by 1.875%

2.2 Practice: Identification (True/False/NEI)

- (a) OLS estimation of demand typically underestimates the price coefficient (in absolute value).
- (b) Using prices of the same product in other geographic markets as an IV is valid because prices in other markets don't affect local demand.
- (c) The logit model solves the dimensionality problem by assuming all products are equally substitutable.

Solutions:

- (a) **TRUE.** Price endogeneity biases α toward zero. Since $\alpha < 0$, this means $|\hat{\alpha}_{OLS}| < |\alpha_{true}|$.
- (b) **TRUE (with caveat).** Hausman IVs work because common cost shocks affect prices in all markets (relevance), but other markets' prices don't directly affect local demand (exclusion). Caveat: requires no common demand shocks.
- (c) **FALSE.** Logit solves dimensionality by using product characteristics. But the IIA problem means substitution is proportional to share, not similarity.

2.3 Practice: Berry Inversion

Question: A market has 3 products plus outside option. Observed shares: $s_{0t} = 0.4$, $s_1 = 0.3$, $s_2 = 0.2$, $s_3 = 0.1$

- (a) Compute the mean utilities δ_1 , δ_2 , δ_3 .
- (b) Rank the products by consumer preference.

Solution: Using $\delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$:

$$\delta_1 = \ln(0.3) - \ln(0.4) = \ln(0.75) \approx -0.29$$

$$\delta_2 = \ln(0.2) - \ln(0.4) = \ln(0.5) \approx -0.69$$

$$\delta_3 = \ln(0.1) - \ln(0.4) = \ln(0.25) \approx -1.39$$

(b) Ranking: Product 1 > Product 2 > Product 3. All products have negative mean utility relative to outside option (which is normalized to $\delta_0 = 0$).

2.4 Practice: Consumer Surplus Change

Question: Market has $\delta_0 = 0$, $\delta_1 = 1$, $\delta_2 = 0.5$. Price coefficient $\alpha = -0.5$. If product 1 is removed from the market, what is the consumer surplus loss per consumer?

Solution:

- **Before removal:**

$$CS^{\text{before}} = \frac{1}{0.5} \ln(e^0 + e^1 + e^{0.5}) = 2 \ln(1 + 2.72 + 1.65) = 2 \ln(5.37) \approx 3.36$$

- **After removal:**

$$CS^{\text{after}} = \frac{1}{0.5} \ln(e^0 + e^{0.5}) = 2 \ln(1 + 1.65) = 2 \ln(2.65) \approx 1.95$$

- **CS Loss:** $3.36 - 1.95 = \$1.41$ per consumer

3 Pricing and Price Discrimination

3.1 Monopoly Pricing: Key Equations

- Profit maximization: $\max_q \pi = p(q) \cdot q - c(q)$
- First-order condition: $MR = MC$
- **Lerner index:**

$$L = \frac{p - MC}{p} = \frac{1}{|\varepsilon|}$$

- **Interpretation:** Markup is inversely related to elasticity
- Equivalently: $p = \frac{MC}{1 - 1/|\varepsilon|}$

3.2 Types of Price Discrimination

1. Perfect price discrimination

- Charge each consumer their exact WTP
- Benchmark; rarely feasible

2. Selection by indicators

- Different prices for observable groups
- Example: student discounts, geographic pricing

3. Self-selection

- Design menu to induce type revelation
- Example: versioning, bundling

3.3 Selection by Indicators: Inverse Elasticity Rule

- Two markets with different elasticities
- Apply Lerner in each market:

$$\frac{p_1 - MC}{p_1} = \frac{1}{|\varepsilon_1|} \quad \text{and} \quad \frac{p_2 - MC}{p_2} = \frac{1}{|\varepsilon_2|}$$

- **Key insight:** Charge higher price in more inelastic market
- Rearranging: $p = \frac{MC}{1 - 1/|\varepsilon|}$

3.4 Two-Part Tariffs

- Structure: $\text{Payment} = F + p \cdot q$
- **Optimal (homogeneous consumers):**
 - Set $p = MC$ (maximize total surplus)
 - Set $F = CS(p = MC)$ (extract all surplus)
- **Result:** Firm captures entire surplus; efficient but inequitable
- **Heterogeneous consumers:** Tradeoff between extraction and participation

3.5 Self-Selection: IC and IR Constraints

- **IC (Incentive Compatibility):** Each type prefers their option

$$\text{IC}_H : \quad v_H^F - p_F \geq v_H^S - p_S$$

- **IR (Individual Rationality):** Each type willing to participate

$$\text{IR}_L : \quad v_L^S - p_S \geq 0$$

- **Optimal menu:** IC binds for H, IR binds for L
- H gets “information rent”; L gets zero surplus

4 Practice Problems: Pricing

4.1 Practice: Lerner Index

Question: A monopolist faces demand $p = 100 - 2q$ and has constant marginal cost $MC = 20$.

- (a) Find the profit-maximizing price and quantity.
- (b) Calculate the Lerner index.
- (c) Verify using the elasticity formula.

Solution:

- (a) $MR = 100 - 4q$. Set $MR = MC$: $100 - 4q = 20 \Rightarrow q = 20$. Price: $p = 100 - 2(20) = 60$
- (b) Lerner index: $L = \frac{60-20}{60} = \frac{40}{60} = \frac{2}{3}$
- (c) Elasticity: $\varepsilon = \frac{dq}{dp} \cdot \frac{p}{q} = -\frac{1}{2} \cdot \frac{60}{20} = -1.5$. Check: $L = \frac{1}{|\varepsilon|} = \frac{1}{1.5} = \frac{2}{3} \checkmark$

4.2 Practice: Selection by Indicators

Question: A firm sells in two markets. Market A has $\varepsilon_A = -3$. Market B has $\varepsilon_B = -5$. Marginal cost is $MC = 10$. Find the optimal price in each market.

Solution: Using $p = \frac{MC}{1-1/|\varepsilon|}$:

- **Market A:** $p_A = \frac{10}{1-1/3} = \frac{10}{2/3} = 15$
- **Market B:** $p_B = \frac{10}{1-1/5} = \frac{10}{4/5} = 12.50$

Higher price in more inelastic market (A).

4.3 Practice: Self-Selection

Question: Two products (Premium, Basic), two consumer types.

	Premium	Basic
High type	100	60
Low type	50	40

$MC = 20$ for both. Equal numbers of each type. A firm considers: $p_P = 100$, $p_B = 40$.

- (a) What does each type buy?
- (b) Is this menu optimal? If not, what should change?

Solution:

- (a) **Consumer choices:**
 - High type: $CS_P = 100 - 100 = 0$, $CS_B = 60 - 40 = 20$. High type buys Basic! (IC violated)
 - Low type: $CS_P = 50 - 100 = -50$, $CS_B = 40 - 40 = 0$. Low type buys Basic
- (b) **Not optimal.** Need to lower p_P so IC binds. IC binds when: $100 - p_P = 60 - 40 = 20 \Rightarrow p_P = 80$. Optimal menu: $p_P = 80$, $p_B = 40$

4.4 Practice: Two-Part Tariff

Question: A gym has demand $q = 20 - p$ from each consumer. Marginal cost is $MC = 4$.

- (a) What is the optimal two-part tariff (F, p) ?
- (b) What is the firm's profit per customer?
- (c) What would happen if firm charged $p = 10$ and $F = 50$?

Solution:

- (a) Optimal: $p = MC = 4$. At $p = 4$: $q = 20 - 4 = 16$. Consumer surplus: $CS = \frac{1}{2}(20 - 4)(16) = 128$. Optimal fee: $F = 128$
- (b) Profit per customer: $F + (p - MC)q = 128 + 0 = 128$
- (c) At $p = 10$: $q = 10$, $CS = \frac{1}{2}(20 - 10)(10) = 50$. $F = 50$ is feasible, but profit = $50 + (10 - 4)(10) = 110 < 128$. Lesson: $p > MC$ leaves money on the table!

4.5 Practice: Backing Out Marginal Cost

Question: Firm observes price $p = 50$ and elasticity $\varepsilon = -2$.

- (a) What is the firm's marginal cost?
- (b) If the elasticity were actually -4 , what would MC be?
- (c) What does this say about the importance of accurate elasticity estimates?

Solution: Using Lerner: $MC = p \left(1 - \frac{1}{|\varepsilon|}\right)$

- (a) $MC = 50(1 - 1/2) = 50(0.5) = 25$
- (b) $MC = 50(1 - 1/4) = 50(0.75) = 37.5$
- (c) Key insight: inferred MC is very sensitive to elasticity! Doubling $|\varepsilon|$ increased inferred MC by 50%. This matters for merger analysis: need accurate elasticities.

5 Key Points Summary

1. **Logit:** $s_{jt} = \exp(\delta_{jt}) / [1 + \sum \exp(\delta_{kt})]$
2. **Berry inversion:** $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$
3. **Elasticities:** Own: $\alpha p_{jt}(1 - s_{jt})$; Cross: $-\alpha p_{kt}s_{kt}$
4. **IVs needed** because $E[p_{jt}\xi_{jt}] \neq 0$; bias toward zero
5. **Lerner:** $L = (p - MC)/p = 1/|\varepsilon|$
6. **Selection by indicators:** Higher price in inelastic market
7. **Two-part tariff:** $p = MC$, $F = CS$
8. **Self-selection:** IC binds for high type, IR for low type